Use-Case Flow Specification

Lab 1 — Problem Statement #35: Customizable Subscription Box Scheduler

Use Case ID / Name	UC-01: Customize Monthly Box
Primary Actor	Subscriber

Preconditions

- The subscriber has an active subscription in good standing.
- The subscriber is authenticated and logged in to the portal.
- The monthly customization window is currently open (i.e., at least 48 hours remain before the billing renewal date, per FR-001).

Postconditions

- The subscriber's monthly box selection is saved and reflects the chosen products and preference tags.
- The updated selection is available to the fulfillment system for inclusion in the next fulfillment manifest (FR-004).
- The subscriber receives on-screen confirmation of the updated box.

Main Success Scenario

1. Subscriber opens the current month's box from their account dashboard.
2. System retrieves and displays the products available for the subscriber's selected preference tags (FR-002).
3. Subscriber selects or swaps products in the box, optionally adjusting preference tags to refine the list.
4. System validates the selection (e.g., quantity limits, product availability, customization window still open).
5. Subscriber reviews and confirms the customized box.
6. System saves the updated selection and timestamps the change.
7. System displays a confirmation message summarizing the finalized box contents.

Alternate Flows

A1. Customization Window Has Closed (branches at step 3)

- 3a1. System detects that fewer than 48 hours remain before the billing renewal date, or the window has already closed.
- 3a2. System prevents further changes to the current cycle's box.
- 3a3. System displays a message: “Customization is closed for this cycle — changes will apply to next month's box.”
- 3a4. Subscriber may choose to edit next month's box instead, or exit the use case.

A2. Selected Product Out of Stock (branches at step 4)

- 4a1. System validation detects that a selected product is unavailable.
- 4a2. System flags the item and suggests an in-stock alternative with a matching preference tag.
- 4a3. Subscriber picks the alternative or removes the item, then resumes at step 4.